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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/123,456	09/06/2018	Shoji KANADA	1982-1042PUS1	7051
2292 7590 03/27/2020 BIRCH STEWART KOLASCH & BIRCH, LLP 8110 Gatehouse Road Suite 100 East Falls Church, VA 22042-1248			EXAMINER BEHNAMIAN, SHAHRIAR	
			ART UNIT	PAPER NUMBER
			2641	
			NOTIFICATION DATE	DELIVERY MODE
			03/27/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

mailroom@bskb.com

Office Action Summary

Application No.

16/123,456

Applicant(s)

KANADA, Shoji

Examiner

SHAHRIAR BEHNAMIAN

Art Unit

2641

AIA (FITF) Status

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 6 September 2018.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-14 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-14 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 6 September 2018 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some** c) ☐ None of the:

1. ☒ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Double Patenting

1. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP §§ 706.02(l)(1) - 706.02(l)(3) for applications not subject to examination under the first inventor

to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

2. Claims 1-14 provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-6 of copending Application No. 16/129,950 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other because they are both directed to system of a learned discriminator acquisition unit that acquires a plurality of learned discriminators obtained by causing each of learning discriminators provided in a plurality of respective terminal devices to perform learning using an image and correct answer data thereof.

3. Claims 1-4, 10-14 provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-15 of copending Application No. 16/129,923 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other because they are both directed to system of a learned discriminator acquisition unit that acquires a plurality of learned discriminators obtained by causing each of learning discriminators provided in a plurality of respective terminal devices to perform learning using an image and correct answer data thereof.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

(a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

4. Claim(s) 1-14 is/are rejected under 35 U.S.C. 102(a)(1)/(a)(2) as being anticipated by Sabe et al. (Publication No. US 2007/0217688).

Regarding claims 1, 9, 10, 11, 13, Sabe teaches a learning assistance device (Sabe, the Abstract), comprising:

a learned discriminator acquisition unit that acquires a plurality of learned discriminators obtained by causing each of learning discriminators provided in a plurality of respective terminal

devices to perform learning using an image and correct answer data thereof (Sabe, pp [118]-[123], [156]-[160]); and

a discriminator output unit that acquires a plurality of discrimination results obtained by causing each of the plurality of learned discriminators to discriminate the same input image, determines correct answer data of the input image on the basis of the plurality of discrimination results and outputs a new learning discriminator obtained by causing the learning discriminator to perform learning again using the input image and the determined correct answer data (Sabe, pp [156]-[160]; Figure 23, pp [235]-[238]);

wherein the learning assistance device repeatedly performs a process in which the learning discriminator acquisition unit acquires the plurality of learned discriminators obtained by causing the each of learning discriminators, the each of learning discriminators being output from the discriminator output unit and being provided for each of the plurality of terminal devices, to perform learning using an image and correct answer data thereof, and the discriminator output unit determines a new correct answer data of a new same input image different from the input image on the basis of a plurality of discrimination results obtained by causing the plurality of learned discriminators acquired by the learned discriminator acquisition unit to discriminate the new same input image, and outputs a new learning discriminator obtained by causing the learning discriminator output by the discriminator output unit to perform learning again using the new input image and the determined new correct answer data (Sabe, Figure 25, pp [302]-[305]; Figure 27, pp [328]-[334], [340]-[341]).

Regarding claim 2, Sabe teaches the discriminator output unit outputs an actually operated discriminator learning the image and the correct answer data thereof used for learning by the new learning discriminator (Sabe, pp [118]-[123], [156]-[160]).

Regarding claim 3, Sabe teaches the learned discriminator acquisition unit acquires the learned discriminator from the plurality of terminal devices over a network, and the discriminator output unit outputs the new learning discriminator to the plurality of terminal devices over the network (Sabe, pp [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claim 4, Sabe teaches the discriminator output unit determines a discrimination result having the largest number of same results among the plurality of discrimination results, as correct answer data of the input image (Sabe, pp [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claim 5, Sabe teaches the discriminator output unit determines a weight of each of the plurality of learned discriminators according to the terminal device learned by the learned discriminator, adds the weights of the learned discriminators having the same result among the discrimination results, and sets a discrimination result having the largest added weight as correct answer data of the input image (Sabe, Figures 5 and 6, pp [16], [22]-[26], [47]-[48], [120]-[122], [143]-[149], [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claim 6, Sabe teaches the discriminator output unit determines the weight of each of the learned discriminators learned at each of the terminal devices according to the number of pieces of correct answer data learned by the learned discriminator at each terminal device, adds the weights of the learned discriminators having the same discrimination result, and sets a discrimination result having the largest added weight as correct answer data of the input image (Sabe, Figures 5 and 6, pp [16], [22]-[26], [47]-[48], [120]-[122], [143]-[149], [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claim 7, Sabe teaches the discriminator output unit determines weights for types of cases of disease of the image learned by the respective learned discriminators with

respect to the respective learned discriminators, adds the weights corresponding to the types of cases of disease of the input image in the learned discriminator having the same discrimination result, and sets a discrimination result having the largest added weight as correct answer data of the input image (Sabe, Figures 5 and 6, pp [16], [22]-[26], [47]-[48], [120]-[122], [143]-[149], [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claim 8, Sabe teaches further comprising: an evaluation unit that evaluates a correct answer rate using an image set including a plurality of images serving as a reference with respect to the plurality of learned discriminators, wherein the discriminator output unit determines the weight of each of the plurality of learned discriminators according to the correct answer rate obtained by the evaluation unit, adds the weights of the respective learned discriminators having the same discrimination results, and sets the discrimination result having the largest added weight as correct answer data of the input image (Sabe, Figures 5 and 6, pp [16], [22]-[26], [47]-[48], [120]-[122], [143]-[149], [156]-[160]; Figure 23, pp [235]-[238]).

Regarding claims 12 and 14, Sabe teaches the discriminator acquisition unit further acquires an actually operated discriminator learning the same image and correct answer data of the image as those of the learning discriminator output from the learning assistance device over a network, and the terminal device further includes a discrimination result acquisition unit that acquires a discrimination result of discriminating an image that is a discrimination target using the actually operated discriminator (Sabe, Figures 5 and 6, pp [16], [22]-[26], [47]-[48], [120]-[122], [143]-[149], [156]-[160]; Figure 23, pp [235]-[238]).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to SHAHRIAR BEHNAMIAN whose telephone number is (571)270-3197. The examiner can normally be reached on Mon-Fri between 7am-7pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Charles Appiah can be reached on (571)272-7904. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/SHAHRIAR BEHNAMIAN/
Primary Examiner, Art Unit 2641